

Nikos Filinis

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Experience

Software Engineer / Research Assistant, NETMODE Lab, NTUA June 2022 – Present

- Developed a multi-cluster orchestration platform using Kubernetes, Karmada and Submariner
- Built backend services for orchestration, observability, and RL-based autoscaling in distributed systems
- Deployed and managed cloud-native applications using Kubernetes, Helm, and Keycloak
- Extended the W3C WoT Python stack with virtual objects, Zenoh and Docker support

Systems Administrator, NETMODE Lab, NTUA June 2022 – Present

- Administered Linux-based infrastructure including virtual machines, networking services and backup systems
- Maintained and troubleshot production services including VPN, DNS, authentication, monitoring and collaboration platforms

Education

PhD - Computer Science Athens, Greece

National Technical University of Athens (NTUA) July 2026 (Expected)

- Dissertation: Orchestration of Distributed Applications across the Computing Continuum (IoT-Edge-Cloud)

Master of Engineering (MEng) - Electrical and Computer Engineering Athens, Greece

National Technical University of Athens (NTUA) July 2021

- Grade: 8.73/10
- Thesis: Autoscaling of serverless applications

Projects

PharmaSea Github

- Built a full-stack crowdsourced medicine price tracking platform using Spring Boot and MySQL
- Implemented REST APIs, user authentication and price submission workflows

PCL Compiler Github

- Implemented a compiler for a Pascal-like language using C++, Flex, Bison and LLVM
- Implemented lexical analysis, parsing, semantic analysis and LLVM IR code generation

Airport Management System Github

- Developed a desktop airport operations management system using JavaFX

Noobcash Github

- Implemented a distributed blockchain system with peer-to-peer communication in Python using Flask
- Developed transaction validation and block generation functionality

Skills

Languages: Java, Python, C++, C, JavaScript, SQL, HTML, CSS

Cloud & Containers: Kubernetes, Karmada, Docker, Helm, Submariner, Cilium

Infrastructure: Linux, OpenStack, VMware ESXi, Proxmox, Networking, DNS, DHCP, VPN

Monitoring & Observability: Prometheus, Grafana, Thanos

Authentication & Security: Keycloak, Authentik, OpenVPN, NetBird

Frameworks & Tools: Spring Boot, Flask, Git, JavaFX, NetBox

Data & Scientific Computing: NumPy, Pandas, Matplotlib

Languages

Greek: Native **English:** C2 **Spanish:** C1 **German:** C1